March 27 - 28

Digital streaming: who upgrades, who waits & who never will

ABA Hackathon Spring ‘26

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Let’s predict when your streaming user will choose short-term savings over long-term value..

Streaming platforms offer tiered subscriptions : ad-supported vs. premium. Every pricing, promotion, and design decision depends on understanding when users prioritize cost savings versus experience quality.

Our Role

Analytics team building predictive models to support tier optimization and customer lifetime value

The Data

1M+ behavioral decisions · 11K+ participants · intertemporal choice experiments with response-time signals

Our Goal

Translate research-grade behavioral data into actionable, interpretable predictions that drive subscription strategy

Dataset & ETL

DATASET AT A GLANCE

1M+
decision observations

11K+
unique participants

Each record contains
— Immediate vs. delayed choice
— Response time (ms)
— Experimental context

DATA SOURCE

Multi-country study about human behavioral decisions:

- where participants traded off an immediate reward against a larger delayed one
- Compiled across multiple lab studies into a single large-scale dataset.

ETL PIPELINE

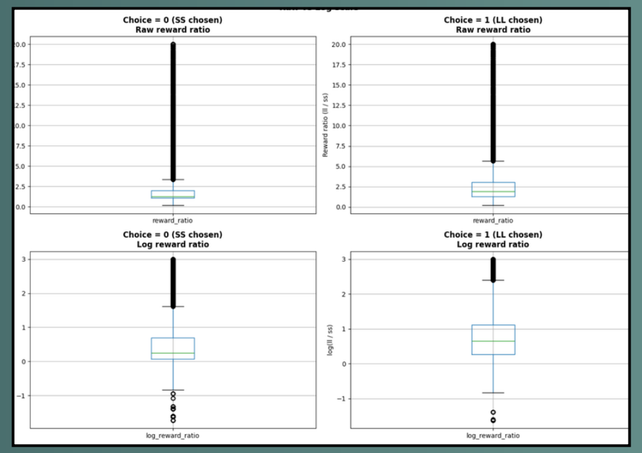
Raw data → Dropped non related → Outlier Handling → Clean dataset

KEY ASSUMPTION

Each observation is treated as independent — participant-level clustering effects are acknowledged but not modeled in this phase. Response time is retained as a behavioral signal, not noise.

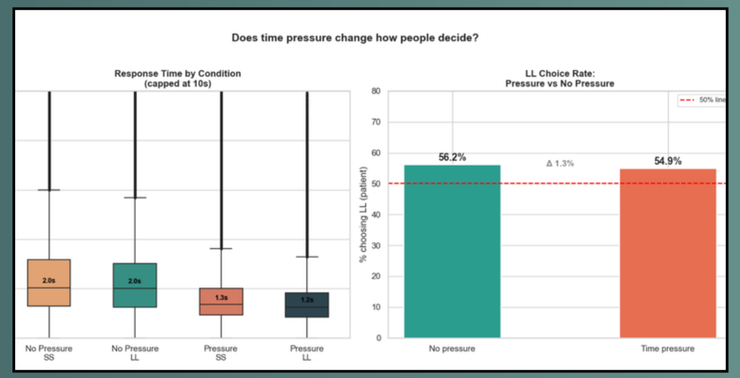
EDA Snapshot

Reward Ratio by Choice



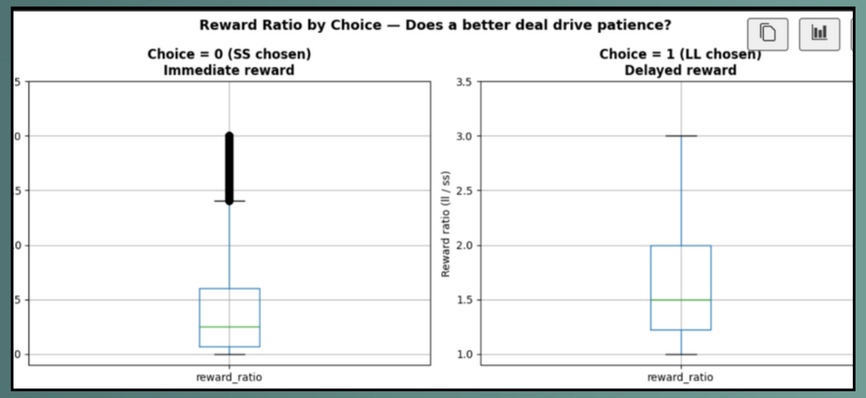
- People choose LL when:
- the bigger-later option offers a better deal
- higher reward ratios associated with more patient choices.

Effect of Time Pressure on Choice



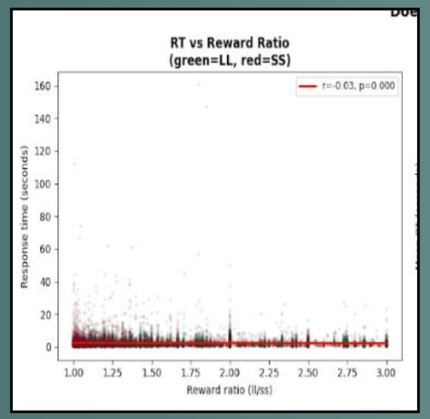
- Time pressure makes people decide much faster
- but it barely changes what they choose
- response times drop sharply under pressure, while LL choice rates remain almost identical.

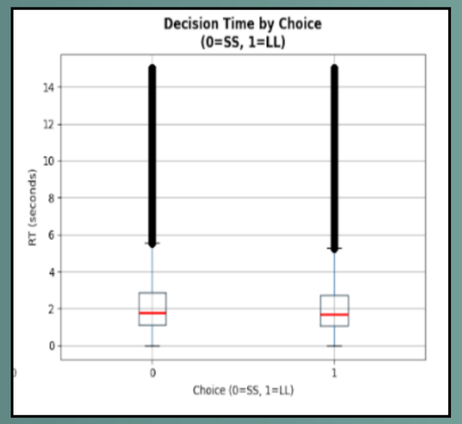
How Reward Payoff Time Influences Choice



- The sooner option is almost always available immediately, while the later option requires waiting from weeks to years — highlighting the stark difference in delay between the two choices.

Response Time vs. Reward Ratio





- Response time barely changes as the reward ratio increases — people don't take longer to decide even when the larger-later option becomes a much better deal.

3 model , 1 target

Target variable: did the user choose the immediate reward? (binary 0/1). Each model was chosen to answer a different question about the same prediction problem.

LOGISTIC REGRESSION

- Logistic Regression draws a straight mathematical line to separate two groups (patient vs. impatient)
- Expect to find that patience_rate strongly pushes people toward LL (patient) while log_discount_k strongly pushes toward SS (impatient)

STRENGTH: INTERPRETABILITY

RANDOM FOREST

- Predict whether a person will choose the patient option or the impatient option Binary 0 or 1
- Train: 80% test: 20%
- Feature interactions that logistic regression misses
- e.g. how response time and reward magnitude interact to influence choice behavior.
- 14 features (raw and engineered)

STRENGTH: NON LINEARITY + ROBUSNESS

XGBOOST

- Gradient boosting on a 1M+ row dataset optimized for predictive accuracy.
- Candidate for production-level scoring of user behavior at scale.

STRENGTH: BEST PREDICTIVE ACCURACY

Result: 3 model , 1 target

Target variable: did the user choose the immediate reward? (binary 0/1). Each model was chosen to answer a different question about the same prediction problem.

LOGISTIC REGRESSION

- Found patience_rate strongly pushes people toward LL (patient) while log_discount_k strongly pushes toward SS (impatient)
- Confirmed by their coefficients (+1.34 and −1.07 respectively)
- ~76% accuracy, slightly weaker than Random Forest because real human behavior isn't perfectly linear.

RANDOM FOREST

```
— RANDOM FOREST —
Accuracy: 0.8223
ROC-AUC:  0.9082
              precision    recall  f1-score   support

           0       0.78      0.83      0.80     83484
           1       0.86      0.82      0.84    105187

    accuracy                           0.82    188671
   macro avg       0.82      0.82      0.82    188671
weighted avg       0.82      0.82      0.82    188671
```

- **key drivers:**
  - **patience_rate (~0.45)** historical customer RT.
  - **log_discount_k (~0.21)** — how steeply someone discounts future rewards.
- SHAP confirms the same story who the person is (patience_rate) matters far more than what the offer is.

XGBOOST

```
— XGBoost Performance —
              precision    recall  f1-score   support

           0       0.73      0.67      0.70     84603
           1       0.76      0.80      0.78    107483

    accuracy                           0.75    192086
   macro avg       0.74      0.74      0.74    192086
weighted avg       0.74      0.75      0.74    192086

AUC: 0.8265
```

- **Key drivers:**
  - **delay_diff** (0.066)
  - **value_ratio** (0.065)
  - **fixed_all_value** (0.047) — When the premium value is anchored and visible, it paradoxically pushes customers away
- Model confirms that customer decisions are non-linear and complex.

STRENGTH: BEST PREDICTIVE ACCURACY

Feature Engineering

VALUE DIFFERENCE (LL-SS)

- Measures how much more valuable the delayed reward is compared to the immediate one
- A direct signal of the premium offer's added benefit

TIME DIFFERENCE IN DAYS (LL DELAY-SS DELAY)

- Captures how much longer the delayed reward requires you to wait compared to the immediate option
- Represents the psychological 'cost of waiting,' a key factor influencing whether users stick with the cheaper tier or consider upgrading

REWARD RATIO (LL VALUE / SS VALUE)

- Measures how attractive the delayed reward is relative to the immediate one
- Helps the model capture whether users respond more to relative value than absolute differences, which is often a stronger driver of upgrade behavior

AGE GROUP

- Age was binned into 12 granular brackets to avoid masking nonlinear behavioral patterns
- This lets the model capture sharp shifts in impatience or upgrade likelihood that a single continuous age variable would smooth out

We should target.. When.. How..

Impulsive user

High k · Fast RT • Chose SS

→ Unlikely to self-upgrade — price-sensitive, short time horizon

→Strategy: time-limited flash offers ("upgrade today, 50% off")

→Focus on ad-tier monetization, not conversion spend

Undecided users

Medium k · Slow RT • Chose SS

→ Slow RT signals genuine deliberation — they're actually considering it

→ Strategy: free trial nudge, social proof, feature comparison

→Highest ROI for personalized engagement spend

Patient users

Low k · Fast RT • Chose LL

→ ~20% of users — naturally long time horizon, high CLV potential

→ Strategy: annual plan messaging, loyalty benefits, value-stacking

→Respond to long-term value framing, not discounts

Remark: **discount rate k**(**k** = (**LL/SS** - **1**) / **delay**) k is the hyperbolic discount rate it measures how impatient someone is when choosing between a reward now vs. later.

However , limitations..

Tipping ratio may not generalize

The 1.29× threshold (LL adoption >50%) is derived from this dataset; real-world customers may respond differently to value ratios.

Task design drives large behavioral differences

Procedure type explains ~30pp variation in LL choices, meaning model predictions are partly shaped by experimental structure rather than pure user traits

Online vs. lab differences complicate interpretation

Online participants show ~8pp more impatience, so results depend on testing environment and may shift in real product contexts

Age effects are nonlinear and hard to generalize

Older users are slower and more variable in RT, but not uniformly less patient — demographic effects are not straightforward

Feature importance reflects task structure

MI ranking shows reward amounts and delays dominate; context variables matter for interpretation but contribute less predictive power

Thresholds and effects vary across subgroups

The tipping ratio, impatience gap, and age-related patterns are not uniform across all demographic or contextual segments, meaning some findings may only hold for specific groups rather than the full population

Recommendations

One-size-fits-all approach won't work!

CUSTOMER SEGMENTATION

Patient Customers

→ Long-term value maximizers

- Respond to: loyalty programs, exclusive perks, tiered rewards, content library depth
- Strategy: High lifetime value (LTV) focus

Impatient Customers

→ Immediate gratification seekers

- Respond to: instant discounts, new content drops, quick wins/achievements
- Strategy: Short-term engagement + conversion to loyalty

→

1. Pricing **subscription tiers** with
   - **upfront value clarity** for patient customers (annual plans, discounts)
   - **flex lower-barrier** entry for impatient customers (free trials, weekly passes).

2. Lifetime Value Management: Use predictive models to identify which segment each customer belongs to based on behavioral signals (churn indicators, engagement patterns)
   - allocate retention budget accordingly.

3. Engagement Strategy:
   - **dual reward** systems loyalty points and exclusive content for patient ones
   - gamification, instant notifications, and limited-time offers for impatient ones.

Thank You
Questions?